SmartHeart: A Machine Learning Approach to Heart Disease Classification

Final Project Report

<u>Presented to</u>

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1. Introduction

Cardiovascular disease (CVD) is one of the leading causes of death worldwide, accounting for millions of fatalities each year. Beyond the human cost, CVD imposes significant financial and operational burdens on healthcare systems through hospital admissions, emergency care, diagnostic tests, and long-term treatment. Clinicians in emergency departments (EDs) and cardiology units face constant pressure to identify high-risk patients and determine who needs urgent care quickly, additional testing, or inpatient admission.

Traditional approaches rely on clinical judgment and scoring systems, such as the Framingham Risk Score. While useful, these methods can be subjective and inconsistent, especially when multiple interacting variables are involved. Machine learning (ML) offers the opportunity to support clinicians by providing data-driven predictions based on historical patient data.

SmartHeart was developed to predict the likelihood of heart disease in patients using a **binary outcome** rather than a detailed severity score. The goal is to answer the practical question: *"Does this patient likely have heart disease?"* By transforming the original UCI Heart Disease dataset, containing 920 patient records, into a binary target variable, SmartHeart provides a consistent second opinion to assist clinicians without replacing their judgment.

2. Dataset Overview and Exploration

2.1 Feature Descriptions

The dataset includes numerical and categorical features reflecting patient demographics, symptoms, clinical measurements, and diagnostic results:

- **Demographics:** Age, sex
- **Symptoms/Risk Indicators:** Chest pain type, fasting blood sugar, exercise-induced angina
- **Clinical Measurements:** Resting blood pressure, cholesterol, maximum heart rate, ST depression (oldpeak)
- **Diagnostic Results:** Resting ECG, ST slope, thallium stress test
- **Target Variables:** Original 5-level severity, binary heart disease outcome

Each feature is clinically meaningful. For example, ST depression indicates stress-induced heart changes, exercise-induced angina reflects reduced coronary blood flow, and abnormal thallium tests identify areas of poor perfusion.

2.2 Data Cleaning and Preprocessing

SmartHeart preprocessing involved:

- Treating physiologically impossible values (e.g., zero cholesterol or blood pressure) as missing
- Converting features to appropriate data types
- Renaming columns for clarity
- Imputing numerical values with the median and categorical values with the mode

These steps ensured a clean, consistent dataset suitable for modeling.

2.3 Exploratory Data Analysis (EDA)

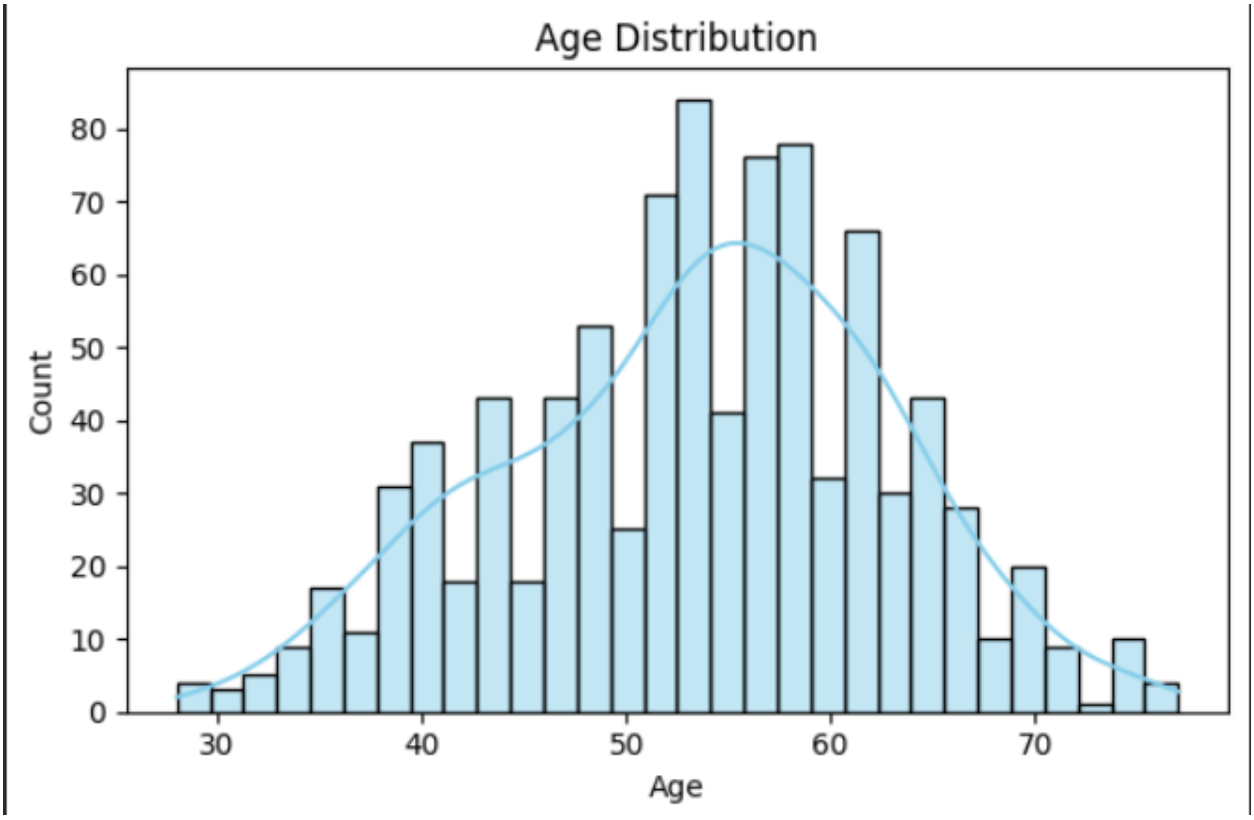
Key findings include:

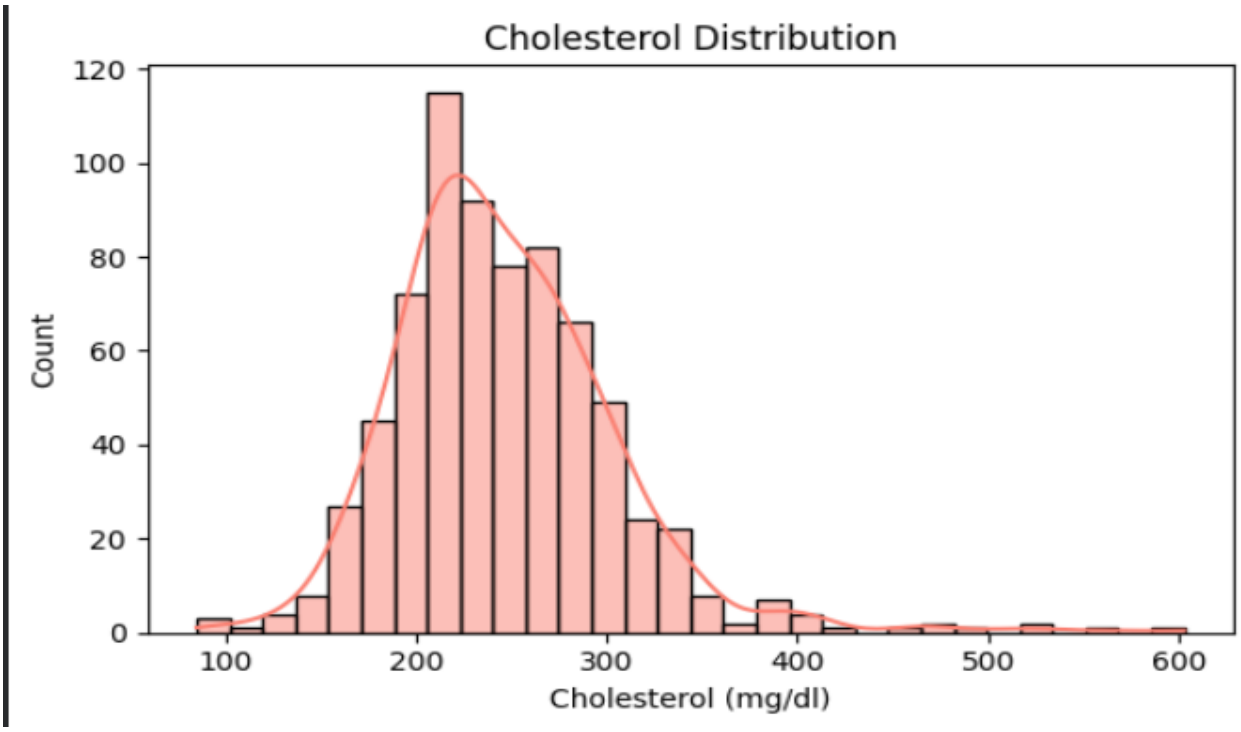
- Most patients were aged 45–65, matching known high-risk age groups

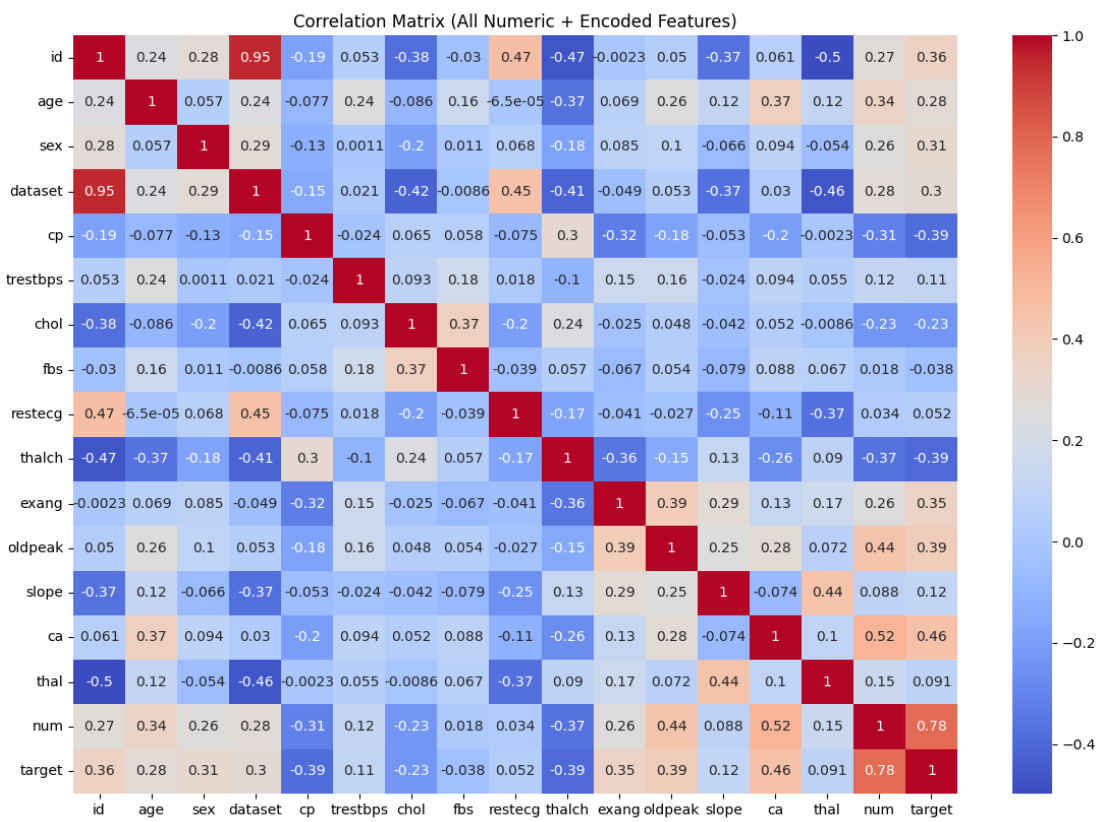
- Blood pressure and cholesterol distributions were right-skewed, indicating outliers
- Higher ST depression and lower maximum heart rate correlated with higher disease risk
- Abnormal thallium tests and asymptomatic chest pain types were more common among patients with heart disease

Correlation analysis revealed:

- Negative correlation between age and maximum heart rate
- Positive correlation between ST depression and disease severity
- No single dominating predictor, indicating multiple interacting features




Cholesterol Distribution
Count
Cholesterol (mg/dl)
0
20
40
60
80
100
120
100
200
300
400
500
600


Correlation Matrix (All Numeric + Encoded Features)
id 1 0.24 0.28 0.95 -0.19 0.053 -0.38 -0.03 0.47 -0.47 -0.0023 0.05 -0.37 0.061 -0.5 0.27 0.36
age 0.24 1 0.057 0.24 -0.077 0.24 -0.086 0.16 -6.5e-05 -0.37 0.069 0.26 0.12 0.37 0.12 0.34 0.28
sex 0.28 0.057 1 0.29 -0.13 0.0011 -0.2 0.011 0.068 -0.18 0.085 0.1 -0.066 0.094 -0.054 0.26 0.31
dataset 0.95 0.24 0.29 1 -0.15 0.021 -0.42 -0.0086 0.45 -0.41 -0.049 0.053 -0.37 0.03 -0.46 0.28 0.3
cp -0.19 -0.077 -0.13 -0.15 1 -0.024 0.065 0.058 -0.075 0.3 -0.32 -0.18 -0.053 -0.2 -0.0023 -0.31 -0.39
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chol -0.38 -0.086 -0.2 -0.42 0.065 0.093 1 0.37 -0.2 0.24 -0.025 0.048 -0.042 0.052 -0.0086 -0.23 -0.23
fbs -0.03 0.16 0.011 -0.0086 0.058 0.18 0.37 1 -0.039 0.057 -0.067 0.054 -0.079 0.088 0.067 0.018 -0.038
restecg 0.47 -6.5e-05 0.068 0.45 -0.075 0.018 -0.2 -0.039 1 -0.17 -0.041 -0.027 -0.25 -0.11 -0.37 0.034 0.052
thalch -0.47 -0.37 -0.18 -0.41 0.3 -0.1 0.24 0.057 -0.17 1 -0.36 -0.15 0.13 -0.26 0.09 -0.37 -0.39
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slope -0.37 0.12 -0.066 -0.37 -0.053 -0.024 -0.042 -0.079 -0.25 0.13 0.29 0.25 1 -0.074 0.44 0.088 0.12
ca 0.061 0.37 0.094 0.03 -0.2 0.094 0.052 0.088 -0.11 -0.26 0.13 0.28 -0.074 1 0.1 0.52 0.46
thal -0.5 0.12 -0.054 -0.46 -0.0023 0.055 -0.0086 0.067 -0.37 0.09 0.17 0.072 0.44 0.1 1 0.15 0.091
num 0.27 0.34 0.26 0.28 -0.31 0.12 -0.23 0.018 0.034 -0.37 0.26 0.44 0.088 0.52 0.15 1 0.78
target 0.36 0.28 0.31 0.3 -0.39 0.11 -0.23 -0.038 0.052 -0.39 0.35 0.39 0.12 0.46 0.091 0.78 1
id age sex dataset cp trestbps chol fbs restecg thalch exang oldpeak slope ca thal num target
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4

3. Methodology

3.1 Problem Definition

The binary outcome for prediction is defined as:

- has_disease = 1 if severity > 0
- has_disease = 0 otherwise

This aligns with clinical workflows and supports threshold-based triage.

3.2 Train/Test Split and Feature Engineering

- 70% training set (n = 644), 30% testing set (n = 276)
- Stratified splitting preserved class proportions
- One-hot encoding resulted in 21 total features
- Numerical features normalized using StandardScaler

3.3 Oversampling Strategies

To address class imbalance, three strategies were tested:

1. No oversampling
2. RandomOverSampler
3. SMOTE (Synthetic Minority Oversampling Technique)

Oversampling was applied **only** to the training set.

3.4 Models Evaluated

- **Logistic Regression:** Interpretable, lightweight, suitable for real-time deployment
- **Decision Tree:** Easy to understand but prone to overfitting
- **K-Nearest Neighbors (k=5):** Captures non-linear relationships, sensitive to noise
- **Random Forest (300 trees):** Robust and accurate, less interpretable

3.5 Evaluation Metrics

- **Accuracy:** Overall correct predictions
- **Precision:** True positives / predicted positives
- **Recall:** True positives / actual positives (critical in healthcare)
- **F1 Score:** Harmonic mean of precision and recall
- **ROC-AUC:** Model's discrimination ability

Recall was prioritized due to the clinical risk of missing a patient with the disease.

3.6 Hyperparameter Tuning

- Logistic Regression: Penalty and regularization
- KNN: Optimized k-value and distance metric
- Decision Tree: Depth and split criteria
- Random Forest: Number of trees and max depth

1. Logistic Regression (Tuned)

Tuning Method: GridSearchCV

Parameters Searched:

- **C:** [0.01, 0.1, 1, 10]
  Regulates the strength of regularization. Lower values impose stronger penalties on large coefficients.

- **Penalty:** ["l1", "l2"]
  L1 encourages sparse models, while L2 provides smoother weight distributions.

Why these parameters?

Logistic Regression serves as the most interpretable baseline model, and tuning focuses on improving its linear boundary without overfitting. Regularization strength and penalty type directly influence how well the model generalizes after SMOTE balancing, ensuring that the model captures meaningful patterns in the features while avoiding unstable or overly complex coefficient estimates.

Outcome:

Tuning produced a model with improved ROC-AUC performance, demonstrating better separation between the positive and negative classes. After tuning, a threshold adjustment using the ROC curve further helped balance sensitivity and specificity for medical decision-making.

2. Decision Tree Classifier (Tuned)

Tuning Method: GridSearchCV

Parameters Searched:

- **max_depth:** [3, 4, 5, None]
  Controls how deep the tree is allowed to grow.

- **min_samples_split:** [2, 5, 10]
  Limits overly specific internal splits.

- **min_samples_leaf:** [1, 2, 4]
  Reduces noise by requiring a minimum number of samples per leaf.

Why these parameters?
Decision Trees are highly flexible and can easily overfit, especially after oversampling. These parameters were chosen to explore how much structure the tree should have and to test whether simpler or more complex branching patterns produce better generalization. Testing both bounded and unbounded depths allowed us to compare stability and interpretability.

Outcome:
The best model used no maximum depth and default leaf sizes, indicating that the tree naturally identified useful splits without needing artificial constraints. The tuned Decision Tree maintained stable performance and remained one of the most interpretable and balanced models in the pipeline.

3. K-Nearest Neighbors (KNN) (Tuned)

Tuning Method: GridSearchCV
Parameters Searched:

- **n_neighbors:** [3, 5, 7, 9]
  Determines how many neighbors are used to classify each sample.

- **weights:** ["uniform", "distance"]
  Allows comparison between equally weighted neighbors and distance-weighted neighbors.

- **p:** [1, 2]
  Selects between Manhattan distance (p=1) and Euclidean distance (p=2).

Why these parameters?
KNN is particularly sensitive to scaling and class distribution. After SMOTE balancing and standardization, tuning focused on selecting a neighborhood size that reduces noise while still capturing relevant local patterns. Evaluating both distance metrics and weighting schemes allowed the model to adapt to the structure of the feature space.

Outcome:
Although tuning improved ROC-AUC and reduced classification noise, KNN remained the weakest overall model. The algorithm struggled with the dataset's dimensionality and mixture of variable types, making it less suitable than the tree-based and linear models.

4. Random Forest Classifier (Tuned)

Tuning Method: RandomizedSearchCV (20 sampled combinations)
Parameters Searched:

- **n_estimators:** [100, 200, 300, 400, 500]

- **max_depth:** [None, 5, 10, 20, 30]

- **min_samples_split:** [2, 5, 10]
- **min_samples_leaf:** [1, 2, 4]
- **max_features:** ["sqrt", "log2"]

Why these parameters?

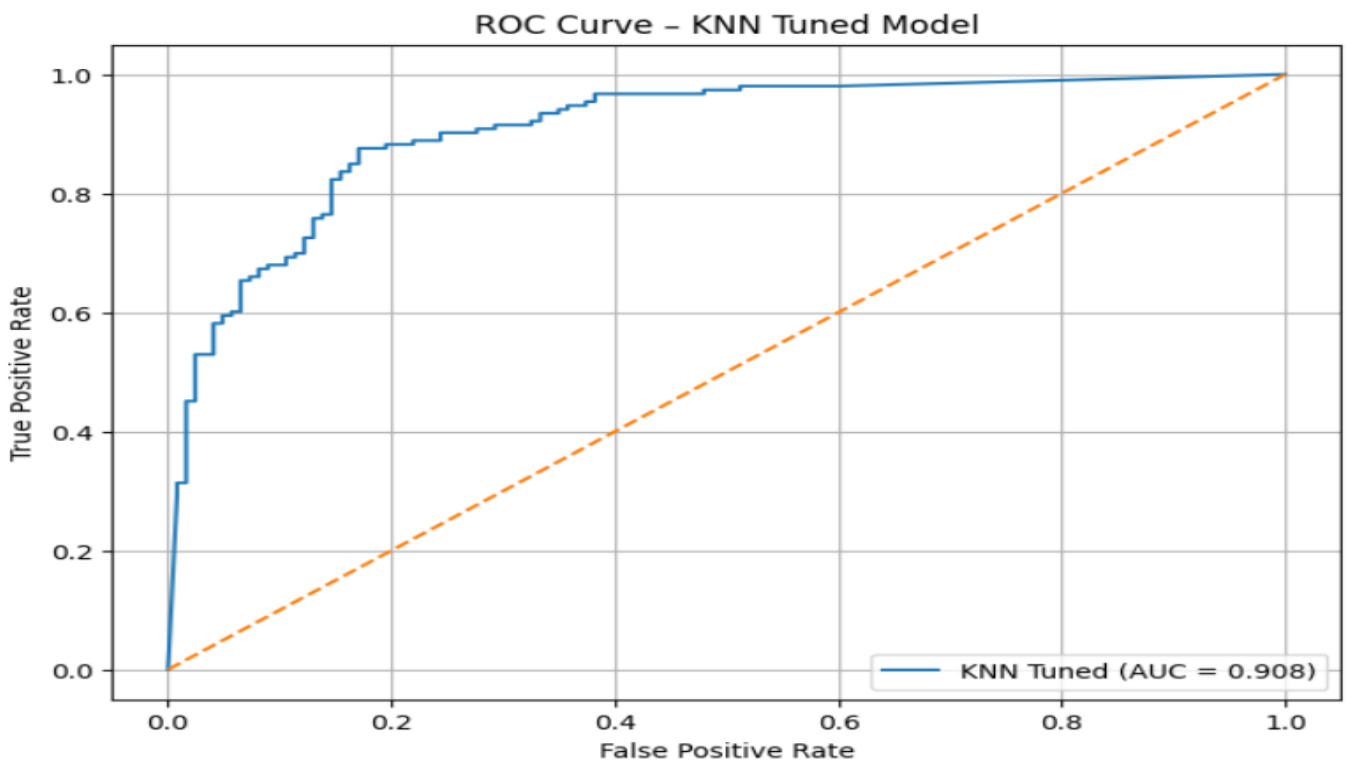
Random Forest benefits from tuning parameters that shape both the individual trees and the ensemble as a whole. These settings allow control over complexity (depth), robustness (minimum split/leaf sizes), and feature randomness (max_features). Together, they help balance bias and variance while ensuring that the model generalizes well after oversampling.

Outcome:

The tuned Random Forest delivered strong ROC-AUC and stable performance but did not outperform the Decision Tree in F1 score for the positive class. This indicates that while the model was highly accurate overall, it relied more heavily on the majority class, reinforcing the importance of F1-based evaluation in imbalanced medical datasets.

3.7 Threshold Optimization

- ROC curve computed and Youden's J statistic evaluated
- Optimal threshold: **0.38**, improving recall from 0.80 → 0.88 and maintaining ROC-AUC ~0.913



4. Results

4.1 Baseline Model Performance

TUNED MODEL SUMMARY

	model	accuracy	precision	recall	f1	roc_auc
0	Logistic Tuned	0.826087	0.830189	0.862745	0.846154	0.910622
2	KNN Tuned	0.851449	0.863636	0.869281	0.866450	0.907806
3	Random Forest Tuned	0.826087	0.852349	0.830065	0.841060	0.902811
1	Decision Tree Tuned	0.778986	0.823944	0.764706	0.793220	0.819358

4.2 Confusion Matrix Analysis

	Predicted No Disease	Predicted Disease
Actual No Disease	150	26
Actual Disease	18	82

4.3 Example Patient Predictions

1. **Patient A:** 55-year-old male, asymptomatic chest pain, ST depression 2.3, abnormal thallium test → predicted has disease (correct)
2. **Patient B:** 47-year-old female, typical angina, high cholesterol, normal ECG → predicted no disease (correct)
3. **Patient C:** 62-year-old male, exercise-induced angina, max heart rate 140 → predicted has disease (correct)

4.4 Feature Importance

- **Risk-increasing:** abnormal thallium test, ST depression, exercise-induced angina
- **Risk-reducing:** higher maximum heart rate, normal ECG

5. Model Insights

- Logistic Regression was chosen for interpretability, high recall, and operational efficiency

- Prioritizing recall reduces false negatives, crucial for patient safety
- Model predictions align with medical knowledge, increasing clinician trust
- SmartHeart supports triage, resource planning, preventive care, and integration into clinical tools

Since the dataset is imbalanced, the evaluation did not rely solely on accuracy. In the current binary setup, the project reports F1 score on the positive "has_disease" class along with ROC-AUC to capture the trade-off between precision (avoiding unnecessary alarms) and recall (avoiding missed disease cases). Conceptually, this follows the same logic often used in multi-class medical problems, where a **macro-averaged F1 score** is preferred: macro F1 gives equal weight to each class rather than allowing the majority class to dominate the metric. If SmartHeart is extended to directly predict all five severity levels (0–4), macro F1 would be a natural primary metric, because it would force the model to perform reasonably well on rarer but clinically critical severe classes, not just on the majority of low-risk patients.

6. Limitations and Future Work

6.1 Limitations

- Modest dataset size (920 records)
- Binary classification loses detailed severity information
- SMOTE introduces synthetic data
- Limited hyperparameter tuning depth
- No temporal or imaging data

6.2 Future Work

- External validation with larger datasets

- Multi-class disease severity prediction
- Integration with EMR/EHR systems
- Probability calibration for risk assessment
- Incorporation of imaging and longitudinal data
- Ethical considerations for fair predictions

7. Conclusion

SmartHeart demonstrates that classical ML techniques, combined with careful preprocessing and threshold optimization, can provide **clinically meaningful, interpretable predictions**. Logistic Regression with SMOTE and ROC-based thresholding offers a strong balance of recall, F1 score, and operational practicality. With further development and integration, SmartHeart can improve patient outcomes, optimize resource allocation, and support early detection of heart disease.

Contributions

Team Member	Role / Contribution
Allen George	Led project development, analyzed the dataset, defined objectives, coordinated research, delegated duties, built the slide deck, performed dataset cleaning, performed EDA analysis, identified key visual diagrams, and finalized the report.
Makayla Pelt	Assisted with the report, made the key findings and conclusions, and participated in project decision discussions.

Ibrahim Idris	Conducted research and wrote the code for the models, dataset cleaning, hyperparameters, and tuning, ensured model performance, key visual diagrams, and participated in project decision discussions.
Sinneh Bangura	Started the research, created the introduction, business impact, and participated in project decision discussions.